Exercise for Sect. 2.1

Exercise **2.1.1**

Show that the space $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}})$ is isomorphic as a measure space to $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2})$.

Proof: For any $x \in \mathbb{T}$, express x in an infinite series $(x_i)_{i\in\mathbb{N}}$ by x_i is the i-th binary place of x. Let $D = \left\{\frac{a}{2^k} : a, k \in \mathbb{N} \text{ and } a < 2^k\right\}$ denote the set of dyadic rationals and D is measure zero since D is countable. Then, define $A^2 \subset \mathbb{T}^2$ by $A^2 = T^2 \setminus (\mathbb{T} \times D \cup D \times \mathbb{T})$. A^2 is measure 1 since $D \times \mathbb{T}$ and $\mathbb{T} \times D$ are measure zero. In this construction, A^2 is the set only containing the points have infinitely many 0 and 1 digits. Define $f :$ $0.x_1x_2x_3\cdots \mapsto (0.x_1x_3x_5\cdots, 0.x_2x_4x_6\cdots)$ and $A = f^{-1}(A^2)$. A is measurable and measure 1 since $A = \mathbb{T} \setminus (D \cup D_{\text{odd}} \cup D_{\text{even}})$ where the two sets stand for the set of points contiains fintely many 0 or 1 in odd or even digits.

To prove f is bimeasurable, define the cylender set $I_{x_1,x_2,\dots,x_n} = \{y \in A : y_i = x_i \text{ for } i \le n\}$ and collection of the intervals $\mathcal{S} \coloneqq \left\{I_{x_1,\dots,x_n} : x_i \in \{0,1\} \text{ and } n \in \mathbb{N}\right\}$ which is a π-system on A that $\sigma(\mathcal{S}) = \mathcal{B}_{\mathbb{T}}$. Similarly, $\mathcal{S}^2 \coloneqq \left\{R_{(x_1,\dots,x_n),(y_1,\dots,y_m)}\right\}$ where $R_{(x_1,\dots,x_n),(y_1,\dots,y_m)} = \left\{(z,w) \in A^2 : z_i = x_i, w_j = y_j \text{ for } i \le n \text{ and } j \le m\right\}$ and $\sigma(\mathcal{S}^2) = \mathcal{B}_{\mathbb{T}^2}$. Then, we want to show f and f^{-1} maps the cylinder sets into measurable sets.

Given an $I_{x_1,x_2,\dots,x_{2k}}$, $f\left(I_{\dots,x_{2k}}\right) = R_{(x_1,\dots,x_{2k-1}),(x_2,\dots,x_{2k})}$, and given $I_{x_1,x_2,\dots,x_{2k+1}}$ get $f(I) = R_{(\dots,x_{2k+1}),(\dots,x_2k)}$. These two situation is trivially measurable.

In contrast, for given $R_{(\dots,x_n),(\dots,y_m)}$, we can divide it into finite disjoint union of cylinders $R_{(\dots,x_k),(\dots,y_k)}$ where $k \le \max(n,m)$. Then the preimage is easy to calculate, $f^{-1}\left(R_{(\dots,x_k),(\dots,y_k)}\right) = I_{x_1,y_1,\dots,x_k,y_k}$ measurable.

Last is to check if f preserves measure. $m_{\mathbb{T}}\left(I_{\dots,x_k}\right) = 2^{-k}$ and $m_{\mathbb{T}^2}\left(R_{(\dots,x_n),(\dots,y_m)}\right) = 2^{-n} \times 2^{-m} = 2^{-nm}$. From the result of mapping above, f and f^{-1} preserves measure on cylinder sets and thus preserves on all measurable sets. Thus, $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}})$ is isomorphic as a measure space to $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2})$. $\square$

Exercise **2.1.2**

Show that the measure-preserving system $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}}, T_4)$, where $T_4(x) = 4x \pmod 1$, is measurably isomorphic to the product system $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2}, T_2 \times T_2)$.

Proof: By the construction above, this is trivial by treating T_k as moving all digit leftward k times. □

Exercise **2.1.3**

For a map $T : X \to X$ and sets $A, B \subseteq X$, prove the following.

- $\chi_A(T(x)) = \chi_{T^{-1}(A)}(x)$;
- $T^{-1}(A \cap B) = T^{-1}(A) \cap T^{-1}(B)$;
- $T^{-1}(A \cup B) = T^{-1}(A) \cup T^{-1}(B)$;
- $T^{-1}(A \Delta B) = T^{-1}(A) \Delta T^{-1}(B)$.

Which of thexe properties also hold with the pre-image under T^{-1} replaced by the forward image under T?

Proof: The four are easy and after replacing, only union still holds. □

Exercise **2.1.4**

Let X be a compact abelian group and let $T : X \to X$ be a countinuous homomorphism. Does T still preserve the Haar measure m_X on X?

Proof: Let T be the constant map from X to identity of X, T satisfies the assumption and may not preserve measure if X contains more than one elements. □

Exercise **2.1.5**

1. Find a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with a non-trivial factor map $\varphi : X \to X$.
2. Find an invertible measure-preserving system $(X, \mathcal{B}, \mu, T)$ with a non-trivial factor map $\varphi : X \to X$.

Proof:

1. $X = \{0, 1\}^{\mathbb{N}}$ be the sample space of a infinite series of outcome of Bernoulli trials, and $\mathcal{B}$ is the σ-algebra generate by all cylinder sets. μ is the Bernulli measure with probability $\frac{1}{2}$ and $\frac{1}{2}$. Define T by $(Tx)_n = x_{n+1}$ for $x \in X$ which is the left shift map on X. Then, claim that $\varphi(x_n) = x_n + x_{n+1} \bmod 2$ is a non-trivial factor map on X to itself.

   First, φ is not an isomorphism since it maps two elements (one solution and its conjugate) to one element, and φ is surjective since for $y \in X, y = y + 0 \bmod 2$. Then, $(\varphi \circ T(x))_n = \varphi\left((Tx)_n\right) = \varphi\left(x_{n+1}\right) = x_{n+1} + x_{n+2} \bmod 2 = (\varphi(x))_{n+1} = T\left((\varphi(x))_n\right) = (T \circ \varphi(x))_n$.

   Last, for C is an cylinder set given by $\left\{x \in X : x_{n_1} = c_1, \ldots, x_{n_k} = c_k\right\}$. If n_i and n_j are not next to each other, the choice of each one won't affect to another one, so we can divides into not adjacency parts and multiply them after computing each part. Thus, we only focus on the situation that $n_j = n_i + 1$, $n_l = n_j + 1$ and so on. If $k = 1$, $\mu\left(\varphi^{-1}(C)\right) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2} = \mu(C)$(11 and 00 or 01 and 10.) If $k = 2$, $\mu\left(\varphi^{-1}(C)\right) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 2 = \frac{1}{4} = \mu(C)$ (the whole process actually depends on the first one.) In conclusion, $\mu\left(\varphi^{-1}(C)\right) = \mu(C)$ for all cylinder sets and thus φ is measure-preserving on $\mathcal{B}$.
2. Just modify X into $\{0, 1\}^{\mathbb{Z}}$, the other settings are still the same.

□

Exercise **2.1.6**

Prove that the circle rotation $R_\alpha : \mathbb{T} \to \mathbb{T}, R_\alpha(t) = t + a_\alpha (\mathrm{mod}\, 1)$ is not measurably isomorphic to the circle-doubling map $T_2 : \mathbb{T} \to \mathbb{T}, T_2(t) = 2t (\mathrm{mod}\, 1)$.

Proof: Assume there is an isomorphic measurable function $\varphi : \mathbb{T} \to \mathbb{T}$ such that $R_\alpha \circ \varphi = \varphi \circ T_2$ almost everywhere. Then, for almost every $x \in \left(0, \frac{1}{2}\right)$ satisfies the equation,

$$\varphi(x) + \alpha \bmod 1 = \varphi(2x)$$

$$(\varphi(2x) - \varphi(x)) \bmod 1 = \alpha$$

$$\Rightarrow \begin{cases} \varphi(2x) - \varphi(a) = \alpha \\ \varphi(2x) - \varphi(b) = \alpha - 1' \end{cases}$$

where one of a, b is x and another is $x + \frac{1}{2}$. Minus that two equations can get $\varphi(b) - \varphi(a) = 1$; however, it is impossible since range of φ is $[0, 1)$. Thus, the two map are not measurable isomorphic. $\square$

Exercise **2.1.7**

Let $\mathfrak{X} = (X, \mathcal{B}, \mu, T)$ be any measure-preserving system. A sub-σ-algebra $\mathcal{A} \subseteq \mathcal{B}_X$ with $T^{-1}\mathcal{A} = \mathcal{A}$ modulo μ is called a T-*invarient sub-σ-algebra*. Show that the system $\tilde{\mathfrak{X}} = \left(\tilde{X}, \tilde{\mathcal{B}}, \tilde{\mu}, \tilde{T}\right)$ defined by

- $\tilde{X} = \left\{x \in X^{\mathbb{Z}} : x_{k+1} = T\left(x_k\right) \text{ for all } k \in \mathbb{Z}\right\}$;
- $\left(\tilde{T}(x)\right)_k = x_{k+1}$ for all $k \in \mathbb{Z}$ and $x \in \tilde{X}$;
- $\tilde{\mu}\left(\left\{x \in \tilde{X} : x_0 \in A\right\}\right) = \mu(A)$ for any $A \in \mathcal{B}$, and $\tilde{\mu}$ is invariant under $\tilde{T}$;
- $\tilde{\mathcal{B}}$ is the smallest $\tilde{T}$-invariant σ-algebra for which the map $\pi : x \mapsto x_0$ from $\tilde{X}$ to X is measurable;

is an invertible measure-preserving system, and that the map $\pi : x \mapsto x_0$ is a factor map. The system $\tilde{\mathfrak{X}}$ is called the invertible extension of $\mathfrak{X}$.

Proof:

- Invertible: $\left(\tilde{T}^{-1}(x)\right)_k = x_{k-1}$ is clearly the inverse map, and $\tilde{T}^{-1}$ is also measurable.
- Measure-preserving: $\tilde{\mu}$ is invarant under $\tilde{T}$ implies the measure-preserving condition.
- Factor map: for any $x \in \tilde{X}$, $\pi \circ \tilde{T}(x) = x_1 = T\left(x_0\right) = T(\pi(x))$.

$\square$

Exercise **2.1.8**

Show that the invertible extension $\tilde{\mathfrak{X}}$ of a measure-preserving system $\mathfrak{X}$ constructed above has the following universal property. For any extension

$$\varphi : (Y, \mathcal{B}_Y, \nu, S) \to (X, \mathcal{B}_X, \mu, T)$$

for which S is invertible, there exists a unique map

$$\tilde{\varphi} : (Y, \mathcal{B}_Y, \nu, S) \to \left(\tilde{X}, \tilde{\mathcal{B}}, \tilde{\mu}, \tilde{T}\right)$$

for which $\varphi = \pi \circ \tilde{\varphi}$.

Proof: Let $\tilde{\varphi}(y_0) = \left(\dots, \varphi\left(S^{-2}y_0\right), \varphi\left(S^{-1}y_0\right), \varphi(y_0), \varphi(Sy_0), \varphi\left(S^2y_0\right), \dots\right)$, and $\tilde{\varphi}$ satisfies $\varphi = \pi \circ \tilde{\varphi}$. Also, $\tilde{\varphi} \circ S = \tilde{T} \circ \tilde{\varphi}$ and $\tilde{\varphi} \circ S^n = \tilde{T^n} \circ \tilde{\varphi}$ for all $n \in \mathbb{Z}$.

If there is another map $\tilde{\varphi}'$ such that $\varphi = \pi \circ \tilde{\varphi}'$, for any $y \in Y$, $\pi \circ \tilde{\varphi}'(y) = \varphi(y) = \pi \circ \tilde{\varphi}(y)$. Using the projection map and the property above,

$$(\tilde{\varphi}'(y))_k = \pi\left(\tilde{T}^k \circ \tilde{\varphi}(y)\right) = \pi(\tilde{\varphi} \circ S^n(y)) = (\tilde{\varphi}(y))_k.$$

Thus, the map is unique. $\square$

Exercise **2.1.9**

1. Show that the invertible extension of the circle-doubling map T_2,
$$X_2 = \left\{x \in \mathbb{T}^{\mathbb{Z}} : x_{k+1} = T_2 x_k \text{ for all } k \in \mathbb{Z}\right\},$$
is a compact abelian group with respect to the coordinate-wise addition defined by $(x+y)_k = x_k + y_k$ for all $k \in \mathbb{Z}$, and the topology inherited from the product topology on $\mathbb{T}^{\mathbb{Z}}$.
2. Show that the diagonal embedding $\delta(r) = (r, r)$ embeds $\mathbb{Z}\left[\frac{1}{2}\right]$ as a descrete subgroup of $\mathbb{R} \times \mathbb{Q}_2$, and that $X_2 \cong \mathbb{R} \times \mathbb{Q}_2 / \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right) \cong \mathbb{R} \times \mathbb{Z}_2 / \delta(\mathbb{Z})$ as compact abelian groups. In particular, the map $\tilde{T}_2$ (which may be thought of as the left shift on X_2, or as the map that doubles in each coordinate) is conjugate to the map
$$(s, r) + \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right) \mapsto (2s, 2r) + \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right)$$
on $\mathbb{R} \times \mathbb{Q}_2 / \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right)$. The group X_2 constructed in this exercise is a simple example of a *solenoid*.

Proof:

1. If $x, y \in X_2$, $(x+y)_k = x_k + y_k = T_2 x_{k-1} + T_2 y_{k-1} = T_2\left(x_{k-1} + y_{k-1}\right) = T_2(x+y)_{k-1}$. The zero sequence 0 is the identity of X_2 and $(x+y)_k = \left(y + x_k\right)$ for all k implies X_2 is an abelian group.

   Since a sequence in a product topology converges if and only if it converges pointwise, the compactness is true by the compactness of each $\mathbb{T}$.

2.

□

Exercise for Sect. 2.2

Exercise **2.2.1**

Prove the following version of Poincare recurrence with a weaker hypothesis (finite additivity in place of countable additivity for the measure) and with a stronger conclusion (a bound on the return time). Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system with μ only assumed to be a finitely additive measure, and let $A \in \mathcal{B}$ have $\mu(A) > 0$. Show that there is some positive $n \leq \frac{1}{\mu(A)}$ for which $\mu(A \cap T^{-n}A) > 0$.

Proof: Suppose $n \in \mathbb{N}$ such that $\mu\left(A \cap T^{-k}(A)\right) = 0$ for all $k \leq n$, then $\mu\left(T^{-j}(A) \cap T^{-k}(A)\right) = 0$ for all $j, k \leq n$ by measure-preserving of T. Therefore,

$$\mu(X) \geq \mu\left(\bigcup_{k=0}^{n} T^{-k}(A)\right) = \sum_{k=1}^{n} T^{-k}(A) = n \cdot \mu(A).$$

The condition of $(X, \mathcal{B}, \mu)$ is a probability space implies the upper bound of n is $\frac{1}{\mu(A)}$.

□

Exercise **2.2.2**

1. If $A \subseteq \mathbb{N}$ has positive density, meaning that

$$d(A) = \lim_{k\to\infty} \frac{1}{k}|A \cap [1,k]|$$

exists and is positive, prove that there is some $n \geq 1$ with $\overline{d}(A \cap (A - n)) > 0$ (here $A - n = \{a - n : a \in A\}$), where

$$\overline{d}(B) = \limsup_{k\to\infty} \frac{1}{k}|B \cap [1,k]|.$$

2. Can you prove this starting with ther weaker assumption that the upper density $\overline{d}(A)$ is positive, and reaching the same conclusion?

Proof:

1. First claim that $\overline{d}$ is a finite additive measure on $\mathbb{P}(\mathbb{N})$.
   - well-defined: for any $B \in \mathbb{P}(\mathbb{N})$, let $d_k = \frac{1}{k}|B \cap [1,k]|$, then $\overline{d}(B) = \limsup d_k$ uniquely exists since d_k is a real sequence.
   - on $[0,1]$: $\overline{d}(\emptyset) = \limsup \frac{1}{k} \cdot 0 = 0$, $d_k \in [0,1]$ for all k implies $\overline{d} : \mathbb{P}(\mathbb{B}) \to [0,1]$.
   - finite additive: if A, B are disjoint, $|(A \cup B) \cap [1,k]| = |A \cap [1,k]| + |B \cap [1,k]|$ for all k implies the finite additivity.

   Then, $\left(\mathbb{N}, \mathbb{P}(\mathbb{N}), \overline{d}\right)$ is a probility space and claim $T(x) = x + 1$ is a measure-preserving map. For all $A \in \mathbb{P}(\mathbb{N})$,

$$\begin{aligned}
\overline{d}\left(T^{-1}(A)\right) &= \limsup_{k\to\infty} \frac{1}{k}|(A-1) \cap [1,k]| \\
&\leq \limsup_{k\to\infty} \frac{1}{k}(|A \cap [1,k]| + 1) \\
&\leq \limsup_{k\to\infty} \frac{1}{k}|A \cap [1,k]| + \limsup_{k\to\infty} \frac{1}{k} \\
&= \limsup_{k\to\infty} \frac{1}{k}|A \cap [1,k]| = \overline{d}(A),
\end{aligned}$$

   and

$$
\begin{aligned}
\overline{d}\left(T^{-1}(A)\right) &= \limsup_{k\to\infty} \frac{1}{k} |(A-1) \cap [1,k]| \\
&\geq \limsup_{k\to\infty} \frac{1}{k} (|A \cap [1,k]| - 1) \\
&\geq \limsup_{k\to\infty} \frac{1}{k} |A \cap [1,k]| - \limsup_{k\to\infty} \frac{1}{k} \\
&= \overline{d}(A).
\end{aligned}
$$

Since A has positive density, $\overline{d}(A) > 0$, then there is an $n \in \mathbb{N}$ satisfies that $\overline{d}(A \cap T^{-n}(A)) > 0$ by the result in 2.2.1.

2. Trivially proved by the arguments above.

$\square$

Exercise **2.2.3**

1. Let (X, d) be a compact metric space and let $T : X \to X$ be a continuous map. Suppose that μ is a T-invariant probability measure defined on the Borel subsets of X. Prove that for μ-almost every $x \in X$ there is a sequence $n_k \to \infty$ with $T^{n_k}(x) \to x$ as $k \to \infty$.
2. Prove that the same conclusion holds under the assumption that X is a metric space, $T : X \to X$ is Borel measurable, and μ is a T-invariant probability measure.

Proof:

1. For all $k \in \mathbb{N}$, let $\cup_{x\in X} D\left(x, \frac{1}{k}\right)$ be the union of open disks on X. Since X is compact, there is finite set I_k such that $\cup_{x\in I} D\left(x, \frac{1}{k}\right)$ cover X. Then, suppose $F_k \subset I_k$ such that $\mu\left(D\left(x, \frac{1}{k}\right)\right) > 0$ for all $x \in I_k \setminus F_k$ and is 0 for all $x \in F_k$. By Poincare's theorem, for $x \in I_k \setminus F_k$ and a.e. $y \in D\left(x, \frac{1}{k}\right)$, $T^n(y) \in D\left(x, \frac{1}{k}\right)$ i.o. Thus, let $G_{x,k} = \left\{y \in D\left(x, \frac{1}{k}\right) : T^n(y) \text{ not back to } D\left(x, \frac{1}{k}\right) \text{ i.o.}\right\}$ and it is measure zero. Union of the sets $D\left(x, \frac{1}{k}\right)$ for $x \in F_k$ and $G_{x,k}$, called E_k, is measure zero since it is union of finite measure zero sets. Last, the union of E_k for all $k \in \mathbb{N}$ is still measure zero and for all $x \in X \setminus \left(\cup E_k\right)$ satisfies $T^{n_k}x \to x$ as $k \to \infty$ by choosing indices from each step.
2. Suppose $S = \text{supp}(\mu) \coloneqq \{x \in X : \mu(D(x, r)) > 0 \quad \forall r > 0\}$ is derived from delete the biggest null open set from the space.

   Prove for the existence of S Let G is a null open set and there is some $x \in X \setminus G$ such that $\mu(D(x, r)) = 0$ for some $r > 0$. Then $G' = G \cup D(x, r) \supset G$ is a null open set. From Zorn's lemma, we can get a maximal element M that is a null open set and $\mu(D(x, r)) > 0$ for all $r > 0$ and $x \in X \setminus M$.

   Then, for any $r > 0$.

   Claim that S is cover by at most countable open ball with radius r Suppose R is a collection of points in S such that $x, y \in R$, $d(x, y) > r$. Then, the set $C =$

$\cup_{x \in R} D\left(x, \frac{r}{2}\right)$ is a union of disjoint sets in R. By the assumption that $D\left(x, \frac{r}{2}\right)$ has positive measure for any $x \in S$, $\frac{r}{2} > 0$, we have $\mu(C) = \sum_{x \in R} \mu\left(D\left(x, \frac{r}{2}\right)\right)$. If R is uncountable, the summation of uncountable positive real numbers goes to infinity, contradicts to the condition of probability space. Thus, R is at most countable and S is cover by $\cup_{x \in \mathbb{R}} D(x, r)$ by finding the biggest R satisfies the assumption.

Let $r = \frac{1}{k}$ for each k, and by the similar prove of 1., we have for almost every $x \in S$ returns to its neighborhood infinitely often with distance at most $2r$. Then take intersection of each k, we have for almost every x, $T^{n_k} x$ converges to x.

$\square$

Exercise for Sect. 2.3

Exercise **2.3.1**

Show that ergodicity is not preserved under direct products as follows. Find a pair of ergodic measure-preserving systems $(X, \mathcal{B}_X, \mu, T)$ and $(Y, \mathcal{B}_Y, \nu, S)$ for which $T \times S$ is not ergodic with respect to the product measure $\mu \times \nu$.

Proof: See exercise 2.3.2. □

Exercise 2.3.2

Define a map $R : \mathbb{T} \times \mathbb{T} \to \mathbb{T} \times \mathbb{T}$ by $R(x, y) = (x + \alpha, y + \alpha)$ for an irrational α. Show that for any set of the form $A \times B$ with A, B measurable subsets of $\mathbb{T}$ (such a set is called a measurable rectangle) has the property of

$$R^{-1}(A \times B) = A \times B \quad \text{(a.e.)} \Rightarrow \left(m_{\mathbb{T}} \times m_{\mathbb{T}}\right)(A \times B) \in \{0, 1\},$$

but the transformation R is not ergodic, even if α is irrational.

Proof: Suppose that $A \times B$ is a rectangle that $R^{-1}(A \times B) = A \times B$ up to measure zero sets. Since R is just a shift map, $R^{-1}(A \times B)$ is still a rectangle and $R^{-1}(A \times B) = R_\alpha^{-1}(A) \times R_\alpha^{-1}(B)$. Thus, we have

$$A = R_\alpha^{-1}(A) \text{ and } B = R_\alpha^{-1}(B) \quad \text{a.e..}$$

By the ergodicity of R_α, we have $m_{\mathbb{T}}(A), m_{\mathbb{T}}(B) \in \{0, 1\}$ and $m_{i(\mathbb{T}^2)}(A \times B) \in \{0, 1\}$.

Besides, we can have an invarient set $I = \left\{(x, y) : |x - y| < \frac{1}{3}\right\}$. The map R is trivially invariant by the criteria of the set. And the measure of the set is $\frac{2}{3}$. □

Exercise 2.3.3

1. Find an arithmetic condition on α_1 and α_2 that is equivalent to the ergodicity of

$R_{\alpha_1} \times R_{\alpha_2} : \mathbb{T} \times \mathbb{T} \to \mathbb{T} \times \mathbb{T}$ with respect to $m_{\mathbb{T}} \times m_{\mathbb{T}}$.

1. Generalize part 1. to characterize ergodicity of the rotation

$$R_{\alpha_1} \times \cdots \times R_{\alpha_n} : \mathbb{T}^n \to \mathbb{T}^n$$

with respect to $m_{\mathbb{T}^n}$.

Proof:

1. Suppose that $T = R_{\alpha_1} \times R_{\alpha_2}$ and f is a T-invariant L^2 function from $\mathbb{T}^2$ to $\mathbb{C}$. By the fourier expansion,

$$f(x, y) = \sum_{n,m \in \mathbb{Z}} c_{n,m} e^{2\pi i(nx+my)}.$$

Thus, by the invariance,

$$\begin{aligned} f(T(x, y)) &= \sum_{n,m \in \mathbb{Z}} c_{n,m} e^{2\pi i\left(n(x+\alpha_1)+m(y+\alpha_2)\right)} \\ &= \sum_{n,m \in \mathbb{Z}} c_{n,m} e^{2\pi i\left(n\alpha_1+m\alpha_2\right)} e^{2\pi i(nx+my)}. \end{aligned}$$

Therefore, T is ergodic if and only if $n\alpha_1 + m\alpha_2 \notin \mathbb{Z}$ for all $n, m \in \mathbb{Z} \setminus \{(0, 0)\}$.

2. Ergodicity is equivalent to $\sum_{i=1}^{n} k_i \alpha_i \notin \mathbb{Z}$ for all $k_i \in \mathbb{Z} \setminus \{0\}$.

$\square$

Exercise 2.3.4

Prove that any factor of an ergodic measure-preserving system is ergodic.

Proof: Suppose $(X, \mathcal{B}, \mu, T)$ is an ergodic system and $(Y, \mathcal{F}, \nu, S)$ is a factor of $(X, \mathcal{B}, \mu, T)$ by the map $\varphi : X \to Y$. Then, if a set $B \in \mathcal{F}$ that $S^{-1}B = B$, let $A = \varphi^{-1}(B)$ and get

$$T^{-1}A = T^{-1} \circ \varphi^{-1}(B) = \varphi^{-1}S^{-1}(B) = \varphi^{-1}(B) = A.$$

Since T is ergodic, $\mu(A) \in \{0, 1\}$. By the measure-preserving property of φ, we have $\nu(B) \in \{0, 1\}$ and S is ergodic. $\square$

Exercise **2.3.5**

Show that for each $p \in [1, \infty]$, a measure-preserving transformation T is ergodic if and only if for any L^p function f, $f \circ T = f$ almost everywhere implies that f is almost everywhere equal to a constant.

Proof: Since $f \circ T = f$ a.e. implies f constant a.e. is equivalent to ergodicity for any f is measurable, thus the ergodicity implies the condition for all $f \in L^p$, $p \in [1, \infty]$. Then, for any B that $B = T^{-1}B$, $\chi_B \in L^p$ for all $p \in [1, \infty]$ and $\chi_B \circ T = \chi_{T^{-1}B} = \chi_B$ implies χ_B is a constant a.e. Therefore, $\mu(B)$ must be in $\{0, 1\}$. $\square$

Exercise **2.3.6**

Show that a measure-preserving transformation T is ergodic if and only if any measurable function $f : X \to \mathbb{R}$ with $f(Tx) \geq f(x)$ almost everywhere is equal to a constant almost everywhere.

Proof: Suppose f measurable and $f(Tx) \geq f(x)$ a.e., then for any $n \in \mathbb{N}, k \in \mathbb{Z}$, define

$$A_n^k = \left\{x \in X : f(x) \leq \frac{k}{n}\right\}.$$

Suppose $A = A_n^k$ that $\mu(A) > 0$. Since $T^{-1}(A) = \left\{x \in X : f(Tx) \leq \frac{k}{n}\right\} \subseteq A$ (a.e.) by $f(Tx) \geq f(x)$ a.e., $\mu(\cup T^{-n}A) = \mu(A)$. By ergodicity of T, $\mu(\cup T^{-n}A) = 1$ if $\mu(A) > 0$ implies $\mu\left(A_n^k\right) \in \{0, 1\}$ for all n, k. Therefore, f can only be constant a.e.

Conversely, suppose $B \subset X$ that $T^{-1}(B) = B$. Then, χ_B is measurable and $\chi_B(Tx) = \chi_{T^{-1}B}(x) = \chi(B)$. Therefore χ_B is constant a.e. implies $\mu(B) \in \{0, 1\}$. $\square$

Exercise **2.3.7**

Let X be a compact metric space and let $T : X \to X$ be continuous. Suppose that μ is a T-invariant ergodic probability measure defined on the Borel subset of X. Prove that for μ-almost every $x \in X$ and every y in the support of μ, there exists a sequence $n_k \nearrow \infty$ such that $T^{n_k}(x) \to y$ as $k \to \infty$. Here the support $\operatorname{supp}(\mu)$ of μ is the smallest closed subset A of X with $\mu(A) = 1$; alternatively

$$\operatorname{supp}(\mu) = X \setminus \bigcup_{\substack{O \subseteq X \text{ open} \\ \mu(O)=0}} O.$$

Notice that X has a countable base for its topology, so the union is still a μ-null set.

Proof: For any $k \in \mathbb{N}$, we can find an finite open cover $\left\{D\left(z_j^k, \frac{1}{2}k\right)\right\}$ of X by the compactness of X. For any $y \in \operatorname{supp}(\mu)$, $y \in D\left(z_j^k, \frac{1}{2}k\right)$ for some $j \in \mathbb{N}$, then $D\left(z_j^k, \frac{1}{2}k\right) \supset D(y, r)$ for some $r > 0$ and thus $\mu\left(D\left(z_j^k, \frac{1}{2}k\right)\right) \geq \mu(D(y, r)) > 0$. Then, let $E_j^k = \left\{x \in X : T^n x \in D\left(z_j^k, \frac{1}{2}k\right) \text{ i.o.}\right\}$, since E_j^k is T-invariant, $\mu\left(E_j^k\right) \in \{0, 1\}$. Since $D\left(z_j^k, \frac{1}{2}k\right)$ has positive measure, by recurrence theorem, almost every points returns to $D\left(z_j^k, \frac{1}{2}k\right)$ implies $\mu\left(E_j^k\right) > 0$. Therefore, let $E_k = \cap_j E_j^k$, E_k has measure one and it goes to somewhere near(with distance less than $\frac{1}{k}$) to any $y \in \operatorname{supp}(\mu)$ infinitely often. Hence, let $E = \cap_k E_k$, we have $\mu(E) = 1$ and for all $x \in E$ and all $y \in \operatorname{supp}(\mu)$, there exists a sequence $n_k \nearrow \infty$ such that $T^{n_k} x \to y$. $\square$

Exercise for Sect. 2.4

Exercise **2.4.1**

Give a different proof that the circle rotation $R_\alpha : \mathbb{T} \to \mathbb{T}$ is ergodic if α is irraional, using Lebesgue's density theorem as follows. Suppose if possible that A and B are measurable invariant sets with $0 < m_{\mathbb{T}}(A), m_{\mathbb{T}}(B) < 1$ and $A \cap B = \varnothing$, and use the fact that the orbit of a point of density for A is dense to show that $A \cap B$ must be non-empty.

Proof: Suppose that α is irrational and A are measurable R_α-invariant set that $m_{\mathbb{T}}(A) \in (0,1)$. Then, let $B = \mathbb{T} \setminus A$, B is R-invariant and $m_{\mathbb{T}}(B) \in (0,1)$.

By the Lebesgue's density theorem, almost every points in A and B is the density point of A or B. For a density point $x \in A, y \in B$ and any constant $c \in (0,1)$, there is an $\varepsilon > 0$ such that

$$m_{\mathbb{T}}(D(x,\varepsilon) \cap A) > 2c\varepsilon$$

$$\text{and } m_{\mathbb{T}}(D(y,\varepsilon) \cap B) > 2c\varepsilon.$$

Since A is R_α-invariant and Lebesgue measure is translation invariant,

$$m_{\mathbb{T}}\left(D\left(R_\alpha^n x, \varepsilon\right) \cap A\right) = m_{\mathbb{T}}(D(x,\varepsilon) \cap A) > 2c\varepsilon.$$

Thus, the points on the orbit of x are density point of A.

Using the fact that α is irrational implies the orbit of any point is dense on $\mathbb{T}$. Then, for the ε, there is an $n \in \mathbb{N}$ such that $\left|R_\alpha^n x - y\right| < \frac{1}{2}\varepsilon$. To simplify the symbols, we rename $R_\alpha^n x$ as x and thus $m_{\mathbb{T}}(D(x,\varepsilon) \cup D(y,\varepsilon)) < \frac{5}{2}\varepsilon$.

By the assumption,

$$\begin{aligned} m_{\mathbb{T}}((D(x,\varepsilon) \cap A) \cup (D(y,\varepsilon) \cap B)) &= m_{\mathbb{T}}(D(x,\varepsilon) \cap A) + m_{\mathbb{T}}(D(y,\varepsilon) \cap B) \\ &> 2c\varepsilon + 2c\varepsilon = 4c\varepsilon. \end{aligned}$$

Since the constant $c \in (0,1)$ is arbitrary, take any $c > \frac{5}{8}$ and get a contradiction by

$$4c\varepsilon < m_{\mathbb{T}}((D(x,\varepsilon) \cap A) \cup (D(y,\varepsilon) \cap B)) \le m_{\mathbb{T}}(D(x,\varepsilon) \cup D(y,\varepsilon)) < \frac{5}{2}\varepsilon.$$

Therefore, any R_α-invariant can only have measure zero or one if α is irrational, and the other side is trivial. $\square$

Exercise 2.4.2

Prove that an ergodic toral automorphism is not measurably isomorphic an ergodic circle rotation.

Proof: Suppose both are the map from $\mathbb{T}^d \to \mathbb{T}^d$. Suppose T is an ergodic (hyperbolic) toral automorphism with the form $T(x) = Ax \bmod 1$, where A is an integer matrix with $\det(A) \neq 0$ and the eigenvalues of A are not on the unit circle. And supppose $R(x) = x + \alpha \bmod 1$ be an ergodic circle rotation, which is, the entries of α is independent on $\mathbb{Q}$.

Let $E_{n(x)} = e^{2\pi i n \cdot x}$ be the basis of Fourier series.

$$U_R(E_n)(x) = e^{2\pi i n \cdot (x+\alpha)} = e^{2\pi i n \cdot \alpha} e^{2\pi i n \cdot x}.$$

Thus, U_R has eigenvalue $e^{2\pi i n \cdot \alpha}$ and eigenfunction $E_n(x)$. However,

$$U_T(E_n)(x) = e^{2\pi i n \cdot Ax} = E_{A^T n}(x),$$

which is, some of E_n is not eigenfunction of U_T.

Assume that the two ergodic system are measurably isomorphic by the map $\varphi : (E_R, \mathcal{B}(\mathbb{T}^d), m_{\mathbb{T}^d}, R) \to (E_T, \mathcal{B}(\mathbb{T}^d), m_{\mathbb{T}^d}, T)$, where $E_R, E_T \subset \mathbb{T}^d$ has full measure. By the assumption that $\varphi \circ T = R \circ \varphi$. Suppose $f, g \in L^2(\mathbb{T}^d)$ such that $g = f \circ \varphi$ and $U_R f = \lambda f$. Then,

$$U_T(g) = g \circ R = f \circ \varphi \circ T = f \circ R \circ \varphi = (U_R f) \circ \varphi.$$

By the assumption that $U_R f = \lambda f$,

$$U_T(g) = \lambda f \circ \varphi = \lambda g,$$

which contradicts to the discussion of eigenfunction of U_T. □

Exercise **2.4.3**

If X is a compact abelian group, prove that the group rotation $R_g(x) = gx$ is ergodic respect to Haar measure if and only if the subgroup $\{g^n : n \in \mathbb{Z}\}$ generate by g is dense in X.

Proof: Let $U_{R_g} : L^2(X) \to L^2(X)$ and $f \in L^2(x)$ is R_g-invariant. Sicne X is a compact abelian group, $\hat{X}$ is an orthogonal basis on $L^2(X)$, and $f = \sum_{\chi\in\hat{X}} c_\chi \chi$. Then, $R_g^k f = \sum_{\chi\in\hat{X}} \left(\chi(g)^k c_\chi\right)\chi$ for all $k \in \mathbb{N}$. Hence we have $c_\chi = \chi(g)^k c_\chi$ for all $\chi \in \hat{X}$ and $k \in \mathbb{N}$.

If $\{g^n\}$ is dense in X, we have $\mathrm{cl}(\{\chi(g)^n\}) = \mathrm{Ran}\,\chi$ and thus $c_\chi = 0$ if $\mathrm{Ran}\,\chi \neq \{1\}$. Therefore, f can only be the trivial character and R_g is ergodic.

If R_g is ergodic, assume that $\{g^n\}$ is not dense in X, which is, $G = \mathrm{cl}(\{g^n\})$ is proper subgroup of X. Then there is a nontrivial character χ that $\chi(y) = 1$ for $y \in G$ and it is R_g-invariant, contradicts to R_g is ergodic.. $\square$

Theorem 0.1 **2.19**

Let $T : X \to X$ be a continuous surjective homomorphism of a compact abelian group X. Then T is ergodic with respect the Haar measure m_X if and only if the identity $\chi(T^n x) = \chi(x)$ for some $n > 0$ and character $\chi \in \hat{X}$ implies that χ is the trivial character with $\chi(x) = 1$ for all $x \in X$.

Corollary 0.2 **2.20**

Let $A \in \mathrm{Mat}_{dd}(\mathbb{Z})$ be an integer matrix with $\det(A) \neq 0$. Then A indices a surjective endomorphism T_A of $\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$ which preserves the Lebesgue measure $m_{\mathbb{T}^d}$. The transformation T_A is ergodic if and only if no eigenvalue of A is a root of unity.

Exercise **2.4.4**

Prove that A is injective if and only if $|\det(A)| = 1$, and general that $A : \mathbb{T}^d \to \mathbb{T}^d$ is $|\det(A)|$-to-one if $\det(A) \neq 0$.

And prove corollary 2.20 using theorem 2.19 and the explicit description of characters on the torus.

Proof: □

Exercise for Sect. 2.5

Exercise **2.5.1**

Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is ergodic if and only if, for any $f, g \in L^2_\mu$,

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \langle U_T^n f, g\rangle = \langle f, 1\rangle \cdot \langle 1, g\rangle.$$

Proof: Suppose T is ergodic, by the continuity of inner product,

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{n-1} \langle U_T^n f, g\rangle = \langle \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{n-1} U_T^n f, g\rangle.$$

By the mean ergodic theorem,

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{n-1} U_T^n f = P_T f,$$

where P_T is the orthogonal projection to the T-invariant subspace. By the ergodicity of T, $P_T f$ is equal to constant c almost everywhere. Since T is measure-preserving,

$$\int_X f \circ T^n \, d\mu = \int_X f \, d\mu$$

$$\Rightarrow c = \langle P_T f, 1\rangle = \int_X \lim_{N\to\infty} U_T^n f \, d\mu = \int f \, d\mu = \langle f, 1\rangle.$$

Thus,

$$\langle \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{n-1} U_T^n f, g\rangle = \langle P_T f, g\rangle = \langle c \cdot 1, g\rangle = c \cdot \langle 1, g\rangle = \langle f, 1\rangle \cdot \langle 1, g\rangle.$$

Now, suppose for any $f, g \in L^2$,

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \langle U_T^n f, g\rangle = \langle f, 1\rangle \cdot \langle 1, g\rangle.$$

Suppose A is a T-invariant set,

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \langle U_T^n \chi_A, \chi_A \rangle = \langle \chi_A, 1 \rangle \cdot \langle 1, \chi_A \rangle = \mu(A)^2.$$

Since A is invariant, the left hand side can be written as $\int \chi_A \chi_A \, \mathrm{d}\mu = \mu(A)$. Therefore, $\mu(A) = \mu(A)^2 \Rightarrow \mu(A) \in \{0, 1\}$ and T is ergodic. $\square$

Exercise 2.5.2

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. For any function f in $L^p_\mu, 1 \le p < \infty$, prove that

$$\frac{1}{n}\sum_{i=0}^{n-1} f(T^i x) \xrightarrow[L^p_\mu]{} f^*,$$

with $f^* \in L^p_\mu$ a T-invariant function.

Proof: Let $U_T : L^p_\mu \to L^p_\mu$ be the Koopman operator defined by $U_T f = f \circ T$. By the measure-preserving property of T, U_T is isometry. Then define $A_N f = \frac{1}{N}\sum_{n=0}^{N-1} U_T^n f$ be the first n average. Since U_T is isometry, $\|A_N\| \le 1$.

For $f \in L^p_\mu$ and any $\varepsilon > 0$, we have a function $g \in L^\infty_\mu$ that $\|f - g\|_p \le \frac{\varepsilon}{3}$. Since $\mu(X) = 1$, $g \in L^2_\mu$ and thus $A_N g \to g'$ by the mean ergodic theorem. Then, need to show $A_N g$ is Cauchy in L^p_μ.

For $p \in [1, 2)$, $\|\cdot\|_p \le \|\cdot\|_2$ (in probability space) implies $A_N g$ is Cauchy in L^p_μ by $A_N g$ is Cauchy in L^2_μ. For $p \in (2.\infty)$, $\|A_N g\|_\infty \le \|g\|_\infty$ and thus $|A_n g - A_m g| \le 2\|g\|_\infty$ almost everywhere. Therefore,

$$\int |A_n g - A_m g|^p \, d\mu = \int |A_n g - A_m g|^{p-2}|A_n g - A_m g|^2 \, d\mu \le \left(2\|g\|_\infty\right)^{p-2}\int |A_n g - A_m g|^2 \, d\mu.$$

Then $\|A_n g - A_m g\|_p \le \left(\left(2\|g\|_\infty\right)^{p-2}\|A_n g\|_2^2\right)^{\frac{1}{p}} \to 0$ as $n, m \to \infty$ by $A_n g$ is Cauchy in L^2_μ.

Thus, choosing N such that for all $n, m \ge N$, $\|A_n g - A_m g\|_p \le \frac{\varepsilon}{3}$ and get

$$\|A_n f - A_m f\| \le \|A_n(f - g)\|_p + \|A_n g - A_m g\|_p + \|A_m(f - g)\|_p \le \varepsilon$$

By the completeness for L^p_μ, we have $A_n f \to f^*$.

Since $U_T(A_n f) = A_n f + \frac{1}{n}(U_T^n f - f)$, $\|U_T(A_n f) - A_n f\|_p \le \frac{1}{n}\left(\|U_T^n f\|_p + \|f\|_p\right) = \frac{2\|f\|_p}{n}$. Thus, $U_T(A_n f) \to f^*$ and we also have $U_T(A_n f) \to U_T f^*$ by continuity of U_T. Therefore, by uniqueness of limit in L^p_μ, f^* is T-invariant. □

Exercise **2.5.3**

Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is ergodic if and only if $A_N(f) \to \int f \, d\mu$ as $N \to \infty$ for all f in a dense subset of L^1_μ.

Proof: If T is ergodic, then just use the ergodic theorem.

Suppose $A_N(f) \to \int f \, d\mu$ as $N \to \infty$ for all f in a dense subset of L^1_μ. Let B be a T-invariant set in $\mathcal{B}$ and thus $\exists f_n \in L^1_\mu$ such that each f_n satisfies $A_N(f_n) \to \int f_n \, d\mu$ and $f_n \to \chi_B$ in L^1_μ.

Since $f_n \to \chi_B$ in L^1_μ, $\int f_n \, d\mu \to \mu(B)$. For any $\varepsilon > 0$, we have an $N_1 \in \mathbb{N}$ such that $\left\| f_n - \chi_B \right\|_1 < \varepsilon$ and $\left| \mu(B) - \int f_n \, d\mu \right| < \varepsilon$ for all $n > N_1$. Then, for all $N \in \mathbb{N}$ and fixed an $n > N_1$,

$$\left\| A_N(f_n) - \chi_B \right\|_1 = \left\| \frac{1}{N} \sum_{i=0}^{N-1} (f_n \circ T^i - \chi_B) \right\|_1 = \frac{1}{N} \left\| \sum_{i=0}^{n-1} (f_n - \chi_B) \circ T^i \right\|_1 \leq \frac{1}{N}(N \cdot \varepsilon) = \varepsilon.$$

Thus, for N is large such that $\left\| A_N(f_n) - \int f_n \, d\mu \right\|_1 < \varepsilon$,

$$\left\| \chi_B - \int \chi_B \, d\mu \right\|_1 \leq \left\| \chi_B - A_N(f_n) \right\|_1 + \left\| A_N(f_n) - \int f_n \, d\mu \right\|_1 + \left\| \int f_n \, d\mu - \int \chi_B \, d\mu \right\|_1 < 3\varepsilon.$$

Since χ_B take values on $\{0, 1\}$, we have $\mu(B) = \int \chi_B \, d\mu \in \{0, 1\}$ and T is ergodic. $\square$

Exercise **2.5.4**

Show that

$$\lim_{N-M\to\infty} \frac{1}{N-M} \sum_{n=M}^{N-1} U_T^n f \to P_T f.$$

Proof: Since U_T is unitary and so continuous,

$$\lim_{N-M\to\infty} \frac{1}{N-M} \sum_{n=M}^{N-1} U_T^n f \underset{k=N-M}{=} U_T^M \left(\lim_{k\to\infty} \frac{1}{k} \sum_{n=0}^{k-1} U_T^n f \right) \to U_T^m (P_T f).$$

Since $P_T f \in I$ which is invariant subspace, $U_T^m P_T f = P_T f$. □

Exercise **2.5.5**

For any set B of positive measure in a measure-preserving system $(X, \mathcal{B}, \mu, T)$,

$$E = \{n \in \mathbb{N} : \mu(B \cap T^{-n}B > 0)\}$$

is syndetic: that is, there are finitely many integers $k_1, \ldots, k_s$ with the property that $\mathbb{N} \subset \bigcup_{i=1}^{s} E - k_i$.

Proof: Suppose B is a positive measure set,

$$\mu(B \cap T^{-n}B) = \int \chi_B \circ T^n \cdot \chi_B \, \mathrm{d}\mu = \langle U_T^n \chi_B, \chi_B \rangle.$$

By 2.5.4, we have

$$\lim_{N-M \to \infty} \frac{1}{N-M} \sum_{n=M}^{N-1} \langle U_T^n \chi_B, \chi_B \rangle = \langle P_T \chi_B, \chi_B \rangle.$$

Since P_T is orthogonal projection, $\langle P_T \chi_B, \chi_B \rangle = \left\| P_T \chi_B \right\|_2^2$. Also, since P_T is orthogonal projection and thus self-adjoint, $\langle P_T \chi_B, \chi_X \rangle = \langle \chi_B, P_T \chi_X \rangle = \langle \chi_B, \chi_X \rangle = \mu(B)$ since χ_X is T-invariant. Then, by Cauchy inequality, $0 < \mu(B)^2 = \left(\langle P_T \chi_B, \chi_X \rangle \right)^2 \leq \left\| P_T \chi_B \right\|_2^2 \cdot \left\| \chi_X \right\|_2^2 = \left\| P_T \chi_B \right\|_2^2$.

For all $\varepsilon > 0$, there is an $L \in \mathbb{N}$ such that for all M, $\left| \frac{1}{N-M} \sum \langle U_T^n \chi_B, \chi_B \rangle - \langle P_T \chi_B, \chi_B \rangle \right| < \varepsilon$ for all $N \geq L$. Take the $\varepsilon = \frac{\mu(B)}{2}$, we have $\frac{1}{L-M} \sum \langle U_T^n \chi_B, \chi_B \rangle > \frac{\mu(B)}{2} > 0$ for any M. Since each term is non-negative, the sum is greater than zero if there is some terms is positive. That is, we have no consecutive L intergers from any start point M is not in E. Thus, take $s = L$ and $k_i = i$ and proved E is syndetic. □

Exercise **2.5.6**

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. We say that T is totally ergodic if T^n is ergodic for all $n \geq 1$. Given $K \geq 1$ define a space $X^{(K)} = X \times \{1, ..., K\}$ with measure $\mu^{(K)} = \mu \times \nu$ defined on the product σ-algebra $\mathcal{B}^{(K)}$, where $\nu(A) = \frac{1}{K}|A|$ is the normalized counting measure defined on any subset $A \subseteq \{1, ..., K\}$, and a $\mu^{(K)}$-preserving transformation $T^{(K)}$ by

$$T^{(K)}(x, i) = \begin{cases} (x, i+1) & \text{if } 1 \leq i < K \\ (Tx, 1) & \text{if } i = K \end{cases}$$

for all $x \in X$. Show that $T^{(K)}$ is ergodic with respect to $\mu^{(K)}$ if and only if T is ergodic with respect to μ, and that $T^{(K)}$ is not totally ergodic if $K > 1$.

Proof: If T is ergodic w.r.t. μ, let $B \in \mathcal{B}^{(K)}$ be an $T^{(K)}$-invariant set. By the invariance, we have if $(x, k) \in B$ for some k, $(x, k) \in B$ for all k and $B|_X$ (B restrict on X) should be T-invariant set. Thus, $\mu^K(B) = \mu\left(B|_X\right) \in \{0, 1\}$ implies the ergodicity of $T^{(K)}$.

If $T^{(K)}$ is ergodic w.r.t. $\mu^{(K)}$, and $B \in \mathcal{B}$ is T-invariant. Then $B^{(K)} = \{(x, k) : x \in B, k \in \{1, ..., k\}\}$ is $T^{(K)}$-invariant with measure $\mu^{(K)}\left(B^{(K)}\right) = \mu(B) \in \{0, 1\}$. Therefore, T is ergodic.

For $K > 1$, $\left(T^{(K)}\right)^K$ is not ergodic with the invariant set $\{(x, 1) : x \in B\}$ for some invariant set B of T and thus $T^{(K)}$ is not totally ergodic. $\square$

Exercise for Sect. 2.6

Exercise **2.6.1**

Let $X = \{x_1, ..., x_r\}$ be a finite set, and let $\sigma : X \to X$ be a permutation of X. The orbit of x_j under σ is the set $\{\sigma^n(x_j)\}_{n \geq 0}$, and σ is called cyclic if there is an orbit of cardinality r.

1. For a cyclic permutation σ and any function $f : X \to \mathbb{R}$, prove that

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} f(\sigma^j x) = \frac{1}{r}(f(x_1) + \cdots + f(x_r)).$$

2. More generally, prove that for any permutation σ and function $f : X \to \mathbb{R}$,

$$\lim_{n\to\infty} \sum_{j=0}^{n-1} f(\sigma^j x) = \frac{1}{p_x}(f(x) + f(\sigma(x))) + \cdots + f(\sigma^{p_x - 1}(x))$$

where the orbit of x has cardinality p_x under σ.

Proof: Let $(X, \mathcal{P}(X), \mu, \sigma)$ be a measure-preserving system by defining $\mu(B) = \frac{1}{r}|B|$ for any set in $\mathcal{P}(X)$. σ is ergodic if it is cyclic.

1. By the ergodic theorem, for any $f : X \to \mathbb{R}$ (it automatically satisfies measurable and L^1), we have $\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} f \circ \sigma^j(x) = \int f(x)\,\mathrm{d}\mu = \frac{1}{r}(f(x_1) + \cdots + f(x_r))$.
2. Just restrict the space to the orbit of x.

□

Exercise **2.6.2**

Minic the proof of finite Vitali covering lemma to prove that for any collection of intervals

$$I_1 = \left[a_1, a_1 + l(1) - 1\right], ..., I_K = \left[a_K, a_K + l(K) - 1\right]$$

in $\mathbb{Z}$ there is a disjoint subcollection $I_{j(1)}, ..., I_{j(k)}$ such that

$$I_1 \cup \cdots \cup I_K \subseteq \bigcup_{m=1}^{k} \left[a_{j(m)} - l(j(m)), a_{j(m)} + 2l(j(M)) - 1\right].$$

Proof: Let $R = \{1, 2, ..., K\}$ be the remaining set. Let $j(1) \in R$ that $l(j(1))$ is the biggest one in $\{l(k) : k \in R\}$. Then, remove the elements e in R such that $I_e \cap I_{j(1)} \neq \emptyset$. We have $I_e \subset \left[a_{j(1)} - l(j(1)), a_{j(1)} + 2l(j(1)) - 1\right]$ since $l(e) \leq l(j(1))$ and $\left|a_e - a_{j(1)}\right| < l(e) + l(j(1)) \leq 2l(j(1))$ by $I_e \cap I_{j(1)} \neq \emptyset$. Then, do the same procedure again and again until there is nothing left in R. We have the disjoint subcollection $I_{j_1}, ..., I_{j_k}$. $\square$

Exercise **2.6.3**

Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system. Prove that, for any $f \in L^1_\mu$,

$$\lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N-1} f(T^n x) = \lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N-1} f(T^{-n} x)$$

almost everywhere.

Proof: By the ergodic theroem, we have $f^*, g^* \in L^1$ such that for almost every $x \in X$,

$$\lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N-1} f(T^n x) = f^*(x), \lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N-1} f(T^{-n} x) = g^*(x).$$

Also we have $\int f \,\mathrm{d}\mu = \int f^* \,\mathrm{d}\mu = \int g^* \,\mathrm{d}\mu$, f^* is T-invariant and g^* is T^{-1}-invariant. Besides, we have $\int f^* - g^* \,\mathrm{d}\mu = 0$ and we want to show that $\int |f^* - g^*| \,\mathrm{d}\mu = 0$.

First, we want to show T-invariance is equivalent to T^{-1}-invariance. Suppose A is a T-invariant set, $T^{-1}A = A$. This implies that $TA = A$ and $TA^c = A^c$. Thus, $\left(T^{-1}\right)^{-1}A = TA = A$ and A is a T^{-1}-invariant set. For a T-invariant L^1 function f, $f \circ T = f$ almost everywhere. Since T is invariable and measure-preserving, compose T^{-1} on both side and get $f = f \circ T^{-1}$ almost everywhere. Therefore, T-invariant function is T^{-1}-invariant.

Next, for any T-invariant set A, restrict the space to A, and use the ergodic theorem again. Then we have $\int_X \chi_A \cdot (f^* - g^*) \,\mathrm{d}\mu = \mu(A) \int_A f^* - g^* \,\mathrm{d}\mu_A = 0$ for any T-invariant set A. (Note that the $\mu_A(B) = \frac{\mu(B)}{\mu(A)}$ is not defined when $\mu(A) = 0$, however, the integral is still zero by the integral on X.)

Then, let $A = \{x \in X : f^*(x) > g^*(x)\}$ and $B = \{x \in X : f^*(x) < g^*(x)\}$. A, B are T-invariant by the invariance of f^* and g^*. By the argument above,

$$\int |f^* - g^*| \,\mathrm{d}\mu = \int_A f^* - g^* \,\mathrm{d}\mu - \int_B f^* - g^* \,\mathrm{d}\mu = 0 + 0 = 0.$$

Thus, $\|f^* - g^*\|_1 = 0$ implies $f^* = g^*$ in L^1 and $f^* = g^*$ almost everywhere. □

Exercise 2.6.4

Fill in the details to prove the estimate in (2.28).

Proof: □

Exercise 2.6.5

Formulate and prove a pointwise ergodic theorem for a measurable function $f \geq 0$ with $\int f \, d\mu = \infty$, under the assumption of ergodicity.

Proof: □